Reproducibility Report: Flores and Coppock (2018)

Do Bilinguals Respond More Favorably to Candidate Advertisements in English or in Spanish?

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This repository holds the actively maintained replication code for Flores and Coppock (2018), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1080/10584609.2018.1426663
Replication archive	10.7910/DVN/XGQ2ZA
Pre-analysis plans	osf.io/e6zjk

The data are not redistributed here. The deposit is 3.7 MB across 14 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

To reproduce. Clone or download the repository, open `flores_coppock_2018.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 14 files, produces every table and figure into `maintained/output/`, and rebuilds the ground truth from what it wrote. Required packages: tidyverse, estimatr, marginaleffects, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about ten seconds once the deposit is on disk. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes. All eight R scripts execute without error on R 4.6.0, with no hardcoded paths beyond a commented instruction to set the working directory and no calls that have since been removed. Every package they name still installs from CRAN, and all of them are alive: `margins` 0.3.28 was released in July 2024, `ggstance` 0.3.7 in April 2024, `coefplot` 1.2.9 in September 2025. The archive's own README omits `margins` from its list of dependencies, which is the only thing standing between a fresh session and a clean run, and it installs in one line.

Running is also reproducing. The ground truth holds 654 rows, 530 of which pair a published value with a value a deposited script prints. Every one of those 530 matches the article to the precision it reports. There are no exceptions: not one coefficient, standard error, sample size, log likelihood or AIC in the article or its appendix disagrees with what the deposited code produces eight years later.

Two things are nonetheless wrong with the deposit, and neither shows up as an error.

Running the archive in place destroys part of it. Both figure scripts end in `ggsave(filename = "flores_coppock_figure_1.pdf", ...)` with a bare relative path, and both of those PDFs are themselves deposited files. A reader who does what the scripts instruct, setting the working directory to the archive folder and sourcing them, overwrites two of the fourteen deposited files and adds a stray `Rplots.pdf`. Nothing about the result changes, since the re-rendered figures carry the same estimates; what changes is that the deposit no longer verifies against its own published checksums.

The README undercounts the archive. It lists seven scripts and describes the archive as containing them. There are eight. `FC_bilinguals_simulation.R`, which draws Figure 3, appears nowhere in it.

1.2 Does the maintained rewrite reproduce the paper?

Every number the article prints reproduces. Of the 630 claims carrying a published value, 624 match it to the precision the article reports. That includes all 47 values of Table 2, which no deposited script computes: 17 the rewrite recovers from the deposited `study_1.csv`, and 30 from the two national surveys the article compares its sample against.

The 6 that do not match are not estimates. They are the endpoints the article gives for two of its five outcome scales, which it describes as running from 1 to 4 where the analysed variables run from 0 to 3. Three further claims are prose that the data contradict: each of three references to a lettered appendix section names the wrong section. All 9 are errors in the article rather than in the code, they move no estimate and no conclusion, and they are corrected in `flores_coppock_2018_errata.pdf` at the root of this repository.

The rewrite contains exactly two numbers not computed from data, `0.05` and `-0.15` in `figure_3_simulation.R`. They are not estimates. They are the supposition the article states on page 18 in order to draw Figure 3: “Suppose for the moment that the positive effect among bilinguals is 5 percentage points but the effect among English-only monolinguals is negative 15 percentage points.” Every other value in `maintained/output/` traces to the deposited data.

2 Paper overview

Citation: Flores, A. and Coppock, A. (2018). “Do bilinguals respond more favorably to candidate advertisements in English or in Spanish?” *Political Communication*, 35(4), 612-633. DOI: 10.1080/10584609.2018.1426663

Summary: Three survey experiments ask whether a candidate gains by speaking Spanish to bilingual voters, and what it costs when the same appeal reaches people who do not speak Spanish. Each uses a real advertisement produced by the campaign in matched English and Spanish versions, so the language of the appeal varies while its content does not: Jeb Bush in the 2016 Republican presidential primary, Filemon Vela running for Congress in Texas, and Mike Coffman running for Congress in Colorado. Experiments 1 and 2 add a second randomized factor, the language of the survey itself, giving a 2 by 2 design among bilinguals; Experiment 3 fields the Vela and Coffman ads to English-speaking monolinguals, where only the ad language varies. Among bilinguals the Spanish ad raises general election support for Bush and for Vela by about 5 percentage points and does nothing for Coffman. Among monolinguals the sign flips and the magnitude grows: support for Coffman falls 18.7 points. Taking the survey in Spanish raises bilinguals’ reported sense of linked fate with other Latinos.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Produces	Status on current R
FC_bilinguals_main_analysis.R	Tables 5 to 9	Clean
FC_bilinguals_in_text_stats.R	Table 3	Clean
FC_bilinguals_interaction_analysis.R	Tables A.1 to A.4	Clean
FC_bilinguals_pid_analysis.R	Tables B.5 to B.8	Clean
FC_bilinguals_logit_analysis.R	Tables C.9 and C.10	Clean once margins is installed; the README does not name it
FC_bilinguals_figure_1.R	Figure 1	Clean; geom_errorbarh() warns that it is deprecated; overwrites a deposited file
FC_bilinguals_figure_2.R	Figure 2	Clean; same deprecation warning; overwrites a deposited file
FC_bilinguals_simulation.R	Figure 3	Clean; absent from the README

Every package the archive names installs from CRAN and every one of them has been updated within the last two years: `stargazer` 5.2.3, `estimatr` 1.0.6, `coefplot` 1.2.9, `ggstance` 0.3.7, `margins` 0.3.28, `tidyverse`, `scales`. The only calls that draw a warning are `geom_errorbarh()` in the two figure scripts, deprecated in ggplot2 4.0.0 in favour of `geom_errorbar(orientation = "y")`, and `ggstance::position_dodgev()`, which warns about overlapping intervals as it always has. Both still draw the figure correctly.

3.1 The archive overwrites itself

The two figure scripts finish with

```
ggsave(filename = "flores_coppock_figure_1.pdf", g, width = 7, height = 5)
```

and `flores_coppock_figure_1.pdf` is one of the fourteen files in the deposit. Running the archive as its own comments instruct therefore replaces two deposited files with fresh renderings and leaves a `Rplots.pdf` behind. Verifying the four affected checksums is the only way to notice, since the output is correct and no error is raised.

Table 2: What a single in-place run of the archive changes.

File	MD5 as deposited	After one run in place
flores_coppock_figure_1.pdf	f382d6ec39e490d6bf47f9259781ea37	differs, and differs again on every run
flores_coppock_figure_2.pdf	a4de3cc99fbe1147ef702f18e21024fe	differs, and differs again on every run
Rplots.pdf	not in the deposit	created

No fixed checksum can be quoted for the overwritten files, because a PDF records the time it was written and so hashes differently on every run. What can be quoted is how far the new rendering is from the old one. The re-rendered figures are not pixel-identical either, since ggplot2's text metrics have moved since 2017; rasterized at 100 dpi, mean absolute pixel intensity differs by 0.9 percent for Figure 1 and 2.4 percent for Figure 2. The difference is confined to glyph rendering and the small shift in panel geometry that follows from it. The plotted estimates are the same. This is damage to the archive's verifiability, not to its result.

`download_original.R` restores the deposit from Dataverse and checks all fourteen files, so the damage is recoverable for anyone who notices it. The lesson generalizes past this archive: run a deposit's scripts in a copy, never in the deposit.

4 Number-by-number comparison

Table 3: Published values by table, against the deposited scripts and against the maintained rewrite.

Location	Claims	In archive output	Archive matches	Archive fails	Rewrite matches	Rewrite fails

table_1	1	0	0	0	0	0
table_2	47	0	0	0	47	0
table_3	16	16	16	0	16	0
table_4	1	0	0	0	0	0
table_5	31	31	31	0	31	0
table_6	31	31	31	0	31	0
table_7	31	31	31	0	31	0
table_8	31	31	31	0	31	0
table_9	10	10	10	0	10	0
figure_1	3	0	0	0	2	0
figure_2	4	0	0	0	3	0
figure_3	2	0	0	0	0	0
table_a1	27	27	27	0	27	0
table_a2	27	27	27	0	27	0
table_a3	27	27	27	0	27	0
table_a4	27	27	27	0	27	0
table_b5	40	40	40	0	40	0
table_b6	40	40	40	0	40	0
table_b7	40	40	40	0	40	0
table_b8	40	40	40	0	40	0
table_c9	41	41	41	0	41	0
table_c10	41	41	41	0	41	0
text	96	30	30	0	72	6

The ground truth records 654 rows: every coefficient, standard error, sample size, log likelihood and AIC in Tables 2, 3 and 5 through 9 and in appendix Tables A.1 through C.10, the sixteen cell counts of Table 3, every quantity the article states in a sentence rather than in a table, and one row for each published float that carries no comparable number, so that a float's absence can never be mistaken for a float that was checked and passed. The typeset tables were extracted from the article and appendix PDFs by parsing rather than by transcription, so that side of the comparison is exhaustive rather than selective; the extraction scripts are in `ground_truth/`.

`value_paper` is carried as the string the page prints, and a value agrees when the pipeline's number, printed to that page's own precision, gives the same digits. The distinction is not cosmetic: 55 of the published values end in a decimal zero that a numeric column silently drops, turning Table 5's standard error of 0.020 into 0.02 and comparing it at a tenth of the intended tolerance.

100 published quantities are not printed by any deposited script: the 47 values of Table 2, discussed below, and 53 quantities the article states in prose, mostly recruitment and quiz-passing counts that the deposit records in its data but never reports.

`ground_truth/build_ground_truth.R` rebuilds the table at the end of every run. Its only hardcoded inputs are the published values, which serve as comparison targets; `value_rewrite` is read back out of `maintained/output/`, so the table cannot drift from the code it grades. `defect_locus` records, for every row that is not a clean match, whether the fault lies with the article, the deposited code, the environment or this repository. 10 rows carry one: 9 are the article, and the remaining 1 is a claim the deposit never computed.

Five published floats carry no number that can be compared against the article, and each has a row saying why.

Table 4: Published floats with no comparable number.

Float	Why there is nothing in it to compare
table_1	Notation rather than results. The table sets out the potential outcomes of bilingual, Spanish-only, English-only and neither-language subjects under the control, English-ad and Spanish-ad conditions. Its eight numeric cells are the two language indicators for each of its four rows; the potential outcomes themselves are symbols.
table_4	The advertisement titles, running times, YouTube links and full transcripts in English and Spanish for all three candidates. Its nine numbers are six running times and three election years, none of which the deposit records.
figure_1	The published figure prints no numbers, so there is nothing in it to compare against the article. All 20 plotted estimates agree with the deposit's own models on all four quantities, to within 0.
figure_2	As Figure 1. All 32 plotted estimates agree with the deposit's own models on all four quantities, to within 0.
figure_3	A calibration exercise rather than an estimate: the net effect of a Spanish-language strategy over the share of bilinguals in the electorate and the risk of mistargeting, holding the two effects at the values the article supposes on page 18. The surface is deterministic given those two numbers, and the rewrite writes 132 of its values to figure_3_simulation.csv.

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity.

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_1	Potential outcomes of four subject types					
table_2	lns, age	33.78			33.78	1
table_2	lns, age_se	0.29			0.2931	1
table_2	lns, cuban	0.04			0.04355	1
table_2	lns, cuban_se	0.00			0.003494	1
table_2	lns, education_5	2.49			2.491	1
table_2	lns, education_5_se	0.02			0.02174	1
table_2	lns, female	0.52			0.5175	1
table_2	lns, female_se	0.01			0.00979	1
table_2	lns, income_7	4.06			4.055	1
table_2	lns, income_7_se	0.04			0.0426	1
table_2	lns, mexican	0.67			0.6667	1
table_2	lns, mexican_se	0.01			0.008874	1
table_2	lns, n	4184			4184	1
table_2	lns, other_hispanic	0.29			0.2897	1
table_2	lns, other_hispanic_se	0.01			0.008515	1
table_2	lucid, age	34.80			34.8	1
table_2	lucid, age_se	0.30			0.3029	1
table_2	lucid, cuban	0.07			0.07411	1
table_2	lucid, cuban_se	0.01			0.006072	1
table_2	lucid, education_5	3.01			3.005	1
table_2	lucid, education_5_se	0.03			0.0251	1
table_2	lucid, female	0.70			0.7003	1
table_2	lucid, female_se	0.01			0.01062	1
table_2	lucid, income_7	4.11			4.109	1
table_2	lucid, income_7_se	0.05			0.05344	1
table_2	lucid, income_9	4.49			4.489	1
table_2	lucid, income_9_se	0.05			0.05204	1
table_2	lucid, mexican	0.49			0.4909	1
table_2	lucid, mexican_se	0.01			0.01159	1
table_2	lucid, n	1862			1862	1
table_2	lucid, other_hispanic	0.44			0.435	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_2	lucid, other_hispanic_se	0.01			0.01149	1
table_2	pew, age	40.11			40.11	1
table_2	pew, age_se	0.84			0.8376	1
table_2	pew, cuban	0.05			0.04662	1
table_2	pew, cuban_se	0.01			0.00852	1
table_2	pew, education_5	2.45			2.452	1
table_2	pew, education_5_se	0.05			0.05378	1
table_2	pew, female	0.46			0.462	1
table_2	pew, female_se	0.02			0.02499	1
table_2	pew, income_9	4.09			4.088	1
table_2	pew, income_9_se	0.11			0.1124	1
table_2	pew, mexican	0.59			0.5941	1
table_2	pew, mexican_se	0.02			0.02403	1
table_2	pew, n	715			715	1
table_2	pew, other_hispanic	0.36			0.3592	1
table_2	pew, other_hispanic_se	0.02			0.0234	1
table_3	experiment_1_bush_bilingual_english_survey, english_ad	462	462	1	462	1
table_3	experiment_1_bush_bilingual_english_survey, spanish_ad	488	488	1	488	1
table_3	experiment_1_bush_bilingual_spanish_survey, english_ad	452	452	1	452	1
table_3	experiment_1_bush_bilingual_spanish_survey, spanish_ad	460	460	1	460	1
table_3	experiment_2_vela_bilingual_english_survey, english_ad	437	437	1	437	1
table_3	experiment_2_vela_bilingual_english_survey, spanish_ad	442	442	1	442	1
table_3	experiment_2_vela_bilingual_spanish_survey, english_ad	420	420	1	420	1
table_3	experiment_2_vela_bilingual_spanish_survey, spanish_ad	382	382	1	382	1
table_3	experiment_2_vela_monolingual_english_survey, english_ad	675	675	1	675	1
table_3	experiment_2_vela_monolingual_english_survey, spanish_ad	669	669	1	669	1
table_3	experiment_3_coffman_bilingual_english_survey, english_ad	455	455	1	455	1
table_3	experiment_3_coffman_bilingual_english_survey, spanish_ad	424	424	1	424	1
table_3	experiment_3_coffman_bilingual_spanish_survey, english_ad	421	421	1	421	1
table_3	experiment_3_coffman_bilingual_spanish_survey, spanish_ad	381	381	1	381	1
table_3	experiment_3_coffman_monolingual_english_survey, english_ad	711	711	1	711	1
table_3	experiment_3_coffman_monolingual_english_survey, spanish_ad	633	633	1	633	1
table_4	Advertisement treatments					
table_5	bush_bilingual, N	1849	1849	1	1849	1
table_5	bush_bilingual, Z_ad	0.049	0.049	1	0.04897	1
table_5	bush_bilingual, Z_ad_se	0.023	0.023	1	0.02322	1
table_5	bush_bilingual, Z_survey	0.005	0.005	1	0.005415	1
table_5	bush_bilingual, Z_survey_se	0.023	0.023	1	0.02323	1
table_5	bush_bilingual, constant	0.449	0.449	1	0.4495	1
table_5	bush_bilingual, constant_se	0.020	0.02	1	0.02009	1
table_5	coffman_bilingual, N	1681	1681	1	1681	1
table_5	coffman_bilingual, Z_ad	0.003	0.003	1	0.002706	1
table_5	coffman_bilingual, Z_ad_se	0.024	0.024	1	0.02362	1
table_5	coffman_bilingual, Z_survey	-0.028	-0.028	1	-0.02779	1
table_5	coffman_bilingual, Z_survey_se	0.024	0.024	1	0.02361	1
table_5	coffman_bilingual, constant	0.384	0.384	1	0.3844	1
table_5	coffman_bilingual, constant_se	0.020	0.02	1	0.02013	1
table_5	coffman_monolingual, N	1343	1343	1	1343	1
table_5	coffman_monolingual, Z_ad	-0.187	-0.187	1	-0.1872	1
table_5	coffman_monolingual, Z_ad_se	0.026	0.026	1	0.02645	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_5	coffman_monolingual, constant	0.513	0.513	1	0.5127	1
table_5	coffman_monolingual, constant_se	0.019	0.019	1	0.01877	1
table_5	vela_bilingual, N	1680	1680	1	1680	1
table_5	vela_bilingual, Z_ad	0.049	0.049	1	0.04855	1
table_5	vela_bilingual, Z_ad_se	0.024	0.024	1	0.0239	1
table_5	vela_bilingual, Z_survey	0.073	0.073	1	0.07296	1
table_5	vela_bilingual, Z_survey_se	0.024	0.024	1	0.0239	1
table_5	vela_bilingual, constant	0.535	0.535	1	0.5348	1
table_5	vela_bilingual, constant_se	0.021	0.021	1	0.02072	1
table_5	vela_monolingual, N	1343	1343	1	1343	1
table_5	vela_monolingual, Z_ad	-0.020	-0.02	1	-0.02012	1
table_5	vela_monolingual, Z_ad_se	0.026	0.026	1	0.02614	1
table_5	vela_monolingual, constant	0.366	0.366	1	0.3659	1
table_5	vela_monolingual, constant_se	0.019	0.019	1	0.01855	1
table_6	bush_bilingual, N	1862	1862	1	1862	1
table_6	bush_bilingual, Z_ad	0.167	0.167	1	0.1668	1
table_6	bush_bilingual, Z_ad_se	0.075	0.075	1	0.07533	1
table_6	bush_bilingual, Z_survey	0.165	0.165	1	0.165	1
table_6	bush_bilingual, Z_survey_se	0.075	0.075	1	0.07546	1
table_6	bush_bilingual, constant	4.810	4.81	1	4.81	1
table_6	bush_bilingual, constant_se	0.063	0.063	1	0.06343	1
table_6	coffman_bilingual, N	1680	1680	1	1680	1
table_6	coffman_bilingual, Z_ad	0.019	0.019	1	0.01945	1
table_6	coffman_bilingual, Z_ad_se	0.077	0.077	1	0.07728	1
table_6	coffman_bilingual, Z_survey	0.093	0.093	1	0.09271	1
table_6	coffman_bilingual, Z_survey_se	0.078	0.078	1	0.07773	1
table_6	coffman_bilingual, constant	4.902	4.902	1	4.902	1
table_6	coffman_bilingual, constant_se	0.066	0.066	1	0.06603	1
table_6	coffman_monolingual, N	1342	1342	1	1342	1
table_6	coffman_monolingual, Z_ad	-0.830	-0.83	1	-0.8299	1
table_6	coffman_monolingual, Z_ad_se	0.074	0.074	1	0.07376	1
table_6	coffman_monolingual, constant	5.161	5.161	1	5.161	1
table_6	coffman_monolingual, constant_se	0.054	0.054	1	0.05444	1
table_6	vela_bilingual, N	1680	1680	1	1680	1
table_6	vela_bilingual, Z_ad	0.068	0.068	1	0.0676	1
table_6	vela_bilingual, Z_ad_se	0.067	0.067	1	0.06747	1
table_6	vela_bilingual, Z_survey	0.149	0.149	1	0.1494	1
table_6	vela_bilingual, Z_survey_se	0.067	0.067	1	0.06745	1
table_6	vela_bilingual, constant	4.994	4.994	1	4.994	1
table_6	vela_bilingual, constant_se	0.057	0.057	1	0.05685	1
table_6	vela_monolingual, N	1341	1341	1	1341	1
table_6	vela_monolingual, Z_ad	-0.472	-0.472	1	-0.4723	1
table_6	vela_monolingual, Z_ad_se	0.070	0.07	1	0.06956	1
table_6	vela_monolingual, constant	4.712	4.712	1	4.712	1
table_6	vela_monolingual, constant_se	0.051	0.051	1	0.05075	1
table_7	bush_bilingual, N	1858	1858	1	1858	1
table_7	bush_bilingual, Z_ad	0.037	0.037	1	0.03737	1
table_7	bush_bilingual, Z_ad_se	0.020	0.02	1	0.02045	1
table_7	bush_bilingual, Z_survey	-0.035	-0.035	1	-0.03475	1
table_7	bush_bilingual, Z_survey_se	0.020	0.02	1	0.02045	1
table_7	bush_bilingual, constant	0.734	0.734	1	0.7342	1
table_7	bush_bilingual, constant_se	0.018	0.018	1	0.01776	1
table_7	coffman_bilingual, N	1680	1680	1	1680	1
table_7	coffman_bilingual, Z_ad	0.035	0.035	1	0.03456	1
table_7	coffman_bilingual, Z_ad_se	0.021	0.021	1	0.02138	1
table_7	coffman_bilingual, Z_survey	-0.045	-0.045	1	-0.04488	1
table_7	coffman_bilingual, Z_survey_se	0.021	0.021	1	0.02148	1
table_7	coffman_bilingual, constant	0.744	0.744	1	0.7442	1
table_7	coffman_bilingual, constant_se	0.018	0.018	1	0.01812	1
table_7	coffman_monolingual, N	1337	1337	1	1337	1
table_7	coffman_monolingual, Z_ad	-0.155	-0.155	1	-0.1552	1
table_7	coffman_monolingual, Z_ad_se	0.024	0.024	1	0.02389	1
table_7	coffman_monolingual, constant	0.815	0.815	1	0.815	1
table_7	coffman_monolingual, constant_se	0.015	0.015	1	0.0146	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_7	vela_bilingual, N	1680	1680	1	1680	1
table_7	vela_bilingual, Z_ad	0.018	0.018	1	0.01761	1
table_7	vela_bilingual, Z_ad_se	0.017	0.017	1	0.01682	1
table_7	vela_bilingual, Z_survey	-0.024	-0.024	1	-0.02399	1
table_7	vela_bilingual, Z_survey_se	0.017	0.017	1	0.01691	1
table_7	vela_bilingual, constant	0.865	0.865	1	0.8647	1
table_7	vela_bilingual, constant_se	0.014	0.014	1	0.01425	1
table_7	vela_monolingual, N	1336	1336	1	1336	1
table_7	vela_monolingual, Z_ad	-0.152	-0.152	1	-0.152	1
table_7	vela_monolingual, Z_ad_se	0.025	0.025	1	0.02466	1
table_7	vela_monolingual, constant	0.780	0.78	1	0.7804	1
table_7	vela_monolingual, constant_se	0.016	0.016	1	0.01596	1
table_8	bush_bilingual, N	1861	1861	1	1861	1
table_8	bush_bilingual, Z_ad	0.076	0.076	1	0.07579	1
table_8	bush_bilingual, Z_ad_se	0.042	0.042	1	0.04152	1
table_8	bush_bilingual, Z_survey	-0.018	-0.018	1	-0.01848	1
table_8	bush_bilingual, Z_survey_se	0.042	0.042	1	0.04156	1
table_8	bush_bilingual, constant	1.768	1.768	1	1.768	1
table_8	bush_bilingual, constant_se	0.035	0.035	1	0.03545	1
table_8	coffman_bilingual, N	1679	1679	1	1679	1
table_8	coffman_bilingual, Z_ad	0.030	0.03	1	0.0304	1
table_8	coffman_bilingual, Z_ad_se	0.041	0.041	1	0.0409	1
table_8	coffman_bilingual, Z_survey	-0.084	-0.084	1	-0.08435	1
table_8	coffman_bilingual, Z_survey_se	0.041	0.041	1	0.04092	1
table_8	coffman_bilingual, constant	1.814	1.814	1	1.814	1
table_8	coffman_bilingual, constant_se	0.034	0.034	1	0.03399	1
table_8	coffman_monolingual, N	1338	1338	1	1338	1
table_8	coffman_monolingual, Z_ad	-0.310	-0.31	1	-0.3101	1
table_8	coffman_monolingual, Z_ad_se	0.044	0.044	1	0.0441	1
table_8	coffman_monolingual, constant	1.966	1.966	1	1.966	1
table_8	coffman_monolingual, constant_se	0.029	0.029	1	0.02947	1
table_8	vela_bilingual, N	1680	1680	1	1680	1
table_8	vela_bilingual, Z_ad	0.103	0.103	1	0.1027	1
table_8	vela_bilingual, Z_ad_se	0.038	0.038	1	0.03787	1
table_8	vela_bilingual, Z_survey	0.011	0.011	1	0.01076	1
table_8	vela_bilingual, Z_survey_se	0.038	0.038	1	0.03793	1
table_8	vela_bilingual, constant	1.915	1.915	1	1.915	1
table_8	vela_bilingual, constant_se	0.033	0.033	1	0.03254	1
table_8	vela_monolingual, N	1338	1338	1	1338	1
table_8	vela_monolingual, Z_ad	-0.160	-0.16	1	-0.1602	1
table_8	vela_monolingual, Z_ad_se	0.047	0.047	1	0.04695	1
table_8	vela_monolingual, constant	1.730	1.73	1	1.73	1
table_8	vela_monolingual, constant_se	0.031	0.031	1	0.03141	1
table_9	bush_bilingual, N	1861	1861	1	1861	1
table_9	bush_bilingual, Z_survey	0.243	0.243	1	0.2427	1
table_9	bush_bilingual, Z_survey_se	0.040	0.04	1	0.0403	1
table_9	bush_bilingual, constant	2.080	2.08	1	2.08	1
table_9	bush_bilingual, constant_se	0.031	0.031	1	0.03121	1
table_9	vela_coffman_bilingual, N	1681	1681	1	1681	1
table_9	vela_coffman_bilingual, Z_survey	0.131	0.131	1	0.1311	1
table_9	vela_coffman_bilingual, Z_survey_se	0.041	0.041	1	0.04127	1
table_9	vela_coffman_bilingual, constant	2.106	2.106	1	2.106	1
table_9	vela_coffman_bilingual, constant_se	0.031	0.031	1	0.03099	1
figure_1	All plotted estimates, standard errors and confidence limits					
figure_1	Estimates Figure 1 plots	20			20	1
figure_1	Figure 1 covers four outcomes	4			4	1
figure_2	All plotted estimates, standard errors and confidence limits					
figure_2	Estimates Figure 2 plots	32			32	1
figure_2	Figure 2 covers four dependent variables	4			4	1
figure_2	Figure 2's caption names four outcomes	4			4	1
figure_3	Net effect surface					
figure_3	Points of the simulated surface the rewrite commits				121	
table_a1	bush_bilingual, N	1849	1849	1	1849	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_a1	bush_bilingual, Z_ad	0.061	0.061	1	0.06065	1
table_a1	bush_bilingual, Z_ad_se	0.032	0.032	1	0.03248	1
table_a1	bush_bilingual, Z_survey	0.018	0.018	1	0.01755	1
table_a1	bush_bilingual, Z_survey_se	0.033	0.033	1	0.03305	1
table_a1	bush_bilingual, constant	0.443	0.443	1	0.4435	1
table_a1	bush_bilingual, constant_se	0.023	0.023	1	0.02319	1
table_a1	bush_bilingual, interaction	-0.024	-0.024	1	-0.02387	1
table_a1	bush_bilingual, interaction_se	0.046	0.046	1	0.04647	1
table_a1	coffman_bilingual, N	1681	1681	1	1681	1
table_a1	coffman_bilingual, Z_ad	-0.039	-0.039	1	-0.03883	1
table_a1	coffman_bilingual, Z_ad_se	0.033	0.033	1	0.03285	1
table_a1	coffman_bilingual, Z_survey	-0.069	-0.069	1	-0.06948	1
table_a1	coffman_bilingual, Z_survey_se	0.033	0.033	1	0.03257	1
table_a1	coffman_bilingual, constant	0.404	0.404	1	0.4044	1
table_a1	coffman_bilingual, constant_se	0.023	0.023	1	0.02303	1
table_a1	coffman_bilingual, interaction	0.087	0.087	1	0.08711	1
table_a1	coffman_bilingual, interaction_se	0.047	0.047	1	0.04724	1
table_a1	vela_bilingual, N	1680	1680	1	1680	1
table_a1	vela_bilingual, Z_ad	0.075	0.075	1	0.07463	1
table_a1	vela_bilingual, Z_ad_se	0.033	0.033	1	0.03346	1
table_a1	vela_bilingual, Z_survey	0.100	0.1	1	0.09969	1
table_a1	vela_bilingual, Z_survey_se	0.034	0.034	1	0.03367	1
table_a1	vela_bilingual, constant	0.522	0.522	1	0.5217	1
table_a1	vela_bilingual, constant_se	0.024	0.024	1	0.02392	1
table_a1	vela_bilingual, interaction	-0.055	-0.055	1	-0.0547	1
table_a1	vela_bilingual, interaction_se	0.048	0.048	1	0.0478	1
table_a2	bush_bilingual, N	1858	1858	1	1858	1
table_a2	bush_bilingual, Z_ad	0.034	0.034	1	0.03409	1
table_a2	bush_bilingual, Z_ad_se	0.028	0.028	1	0.02803	1
table_a2	bush_bilingual, Z_survey	-0.038	-0.038	1	-0.03815	1
table_a2	bush_bilingual, Z_survey_se	0.030	0.03	1	0.02985	1
table_a2	bush_bilingual, constant	0.736	0.736	1	0.7359	1
table_a2	bush_bilingual, constant_se	0.021	0.021	1	0.02053	1
table_a2	bush_bilingual, interaction	0.007	0.007	1	0.006695	1
table_a2	bush_bilingual, interaction_se	0.041	0.041	1	0.04095	1
table_a2	coffman_bilingual, N	1680	1680	1	1680	1
table_a2	coffman_bilingual, Z_ad	0.051	0.051	1	0.05097	1
table_a2	coffman_bilingual, Z_ad_se	0.029	0.029	1	0.02872	1
table_a2	coffman_bilingual, Z_survey	-0.028	-0.028	1	-0.02843	1
table_a2	coffman_bilingual, Z_survey_se	0.030	0.03	1	0.03033	1
table_a2	coffman_bilingual, constant	0.736	0.736	1	0.7363	1
table_a2	coffman_bilingual, constant_se	0.021	0.021	1	0.02068	1
table_a2	coffman_bilingual, interaction	-0.034	-0.034	1	-0.0344	1
table_a2	coffman_bilingual, interaction_se	0.043	0.043	1	0.04292	1
table_a2	vela_bilingual, N	1680	1680	1	1680	1
table_a2	vela_bilingual, Z_ad	0.013	0.013	1	0.01254	1
table_a2	vela_bilingual, Z_ad_se	0.022	0.022	1	0.02246	1
table_a2	vela_bilingual, Z_survey	-0.029	-0.029	1	-0.02918	1
table_a2	vela_bilingual, Z_survey_se	0.024	0.024	1	0.02425	1
table_a2	vela_bilingual, constant	0.867	0.867	1	0.8673	1
table_a2	vela_bilingual, constant_se	0.016	0.016	1	0.01625	1
table_a2	vela_bilingual, interaction	0.011	0.011	1	0.01062	1
table_a2	vela_bilingual, interaction_se	0.034	0.034	1	0.03379	1
table_a3	bush_bilingual, N	1861	1861	1	1861	1
table_a3	bush_bilingual, Z_ad	0.135	0.135	1	0.1349	1
table_a3	bush_bilingual, Z_ad_se	0.058	0.058	1	0.05766	1
table_a3	bush_bilingual, Z_survey	0.043	0.043	1	0.04288	1
table_a3	bush_bilingual, Z_survey_se	0.058	0.058	1	0.05816	1
table_a3	bush_bilingual, constant	1.738	1.738	1	1.738	1
table_a3	bush_bilingual, constant_se	0.041	0.041	1	0.04061	1
table_a3	bush_bilingual, interaction	-0.121	-0.121	1	-0.1206	1
table_a3	bush_bilingual, interaction_se	0.083	0.083	1	0.08305	1
table_a3	coffman_bilingual, N	1679	1679	1	1679	1
table_a3	coffman_bilingual, Z_ad	-0.040	-0.04	1	-0.03951	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_a3	coffman_bilingual, Z_ad_se	0.056	0.056	1	0.0562	1
table_a3	coffman_bilingual, Z_survey	-0.154	-0.154	1	-0.1544	1
table_a3	coffman_bilingual, Z_survey_se	0.057	0.057	1	0.05654	1
table_a3	coffman_bilingual, constant	1.848	1.848	1	1.848	1
table_a3	coffman_bilingual, constant_se	0.038	0.038	1	0.03848	1
table_a3	coffman_bilingual, interaction	0.146	0.146	1	0.1464	1
table_a3	coffman_bilingual, interaction_se	0.082	0.082	1	0.08188	1
table_a3	vela_bilingual, N	1680	1680	1	1680	1
table_a3	vela_bilingual, Z_ad	0.107	0.107	1	0.1074	1
table_a3	vela_bilingual, Z_ad_se	0.052	0.052	1	0.05209	1
table_a3	vela_bilingual, Z_survey	0.016	0.016	1	0.01553	1
table_a3	vela_bilingual, Z_survey_se	0.054	0.054	1	0.05353	1
table_a3	vela_bilingual, constant	1.913	1.913	1	1.913	1
table_a3	vela_bilingual, constant_se	0.037	0.037	1	0.03741	1
table_a3	vela_bilingual, interaction	-0.010	-0.01	1	-0.009758	1
table_a3	vela_bilingual, interaction_se	0.076	0.076	1	0.07587	1
table_a4	bush_bilingual, N	1862	1862	1	1862	1
table_a4	bush_bilingual, Z_ad	0.248	0.248	1	0.248	1
table_a4	bush_bilingual, Z_ad_se	0.101	0.101	1	0.1014	1
table_a4	bush_bilingual, Z_survey	0.249	0.249	1	0.2493	1
table_a4	bush_bilingual, Z_survey_se	0.107	0.107	1	0.1072	1
table_a4	bush_bilingual, constant	4.768	4.768	1	4.768	1
table_a4	bush_bilingual, constant_se	0.072	0.072	1	0.07207	1
table_a4	bush_bilingual, interaction	-0.166	-0.166	1	-0.1657	1
table_a4	bush_bilingual, interaction_se	0.151	0.151	1	0.1509	1
table_a4	coffman_bilingual, N	1680	1680	1	1680	1
table_a4	coffman_bilingual, Z_ad	-0.089	-0.089	1	-0.0886	1
table_a4	coffman_bilingual, Z_ad_se	0.104	0.104	1	0.1036	1
table_a4	coffman_bilingual, Z_survey	-0.016	-0.016	1	-0.0156	1
table_a4	coffman_bilingual, Z_survey_se	0.111	0.111	1	0.1108	1
table_a4	coffman_bilingual, constant	4.954	4.954	1	4.954	1
table_a4	coffman_bilingual, constant_se	0.076	0.076	1	0.07558	1
table_a4	coffman_bilingual, interaction	0.226	0.226	1	0.2265	1
table_a4	coffman_bilingual, interaction_se	0.155	0.155	1	0.1551	1
table_a4	vela_bilingual, N	1680	1680	1	1680	1
table_a4	vela_bilingual, Z_ad	0.166	0.166	1	0.1658	1
table_a4	vela_bilingual, Z_ad_se	0.093	0.093	1	0.09341	1
table_a4	vela_bilingual, Z_survey	0.250	0.25	1	0.2502	1
table_a4	vela_bilingual, Z_survey_se	0.092	0.092	1	0.0925	1
table_a4	vela_bilingual, constant	4.945	4.945	1	4.945	1
table_a4	vela_bilingual, constant_se	0.065	0.065	1	0.06474	1
table_a4	vela_bilingual, interaction	-0.206	-0.206	1	-0.2062	1
table_a4	vela_bilingual, interaction_se	0.135	0.135	1	0.135	1
table_b5	coffman_bilingual_democrat, N	1029	1029	1	1029	1
table_b5	coffman_bilingual_democrat, Z_ad	-0.003	-0.003	1	-0.002698	1
table_b5	coffman_bilingual_democrat, Z_ad_se	0.027	0.027	1	0.02693	1
table_b5	coffman_bilingual_democrat, constant	0.248	0.248	1	0.2481	1
table_b5	coffman_bilingual_democrat, constant_se	0.019	0.019	1	0.01867	1
table_b5	coffman_bilingual_republican, N	335	335	1	335	1
table_b5	coffman_bilingual_republican, Z_ad	0.022	0.022	1	0.02232	1
table_b5	coffman_bilingual_republican, Z_ad_se	0.045	0.045	1	0.0452	1
table_b5	coffman_bilingual_republican, constant	0.771	0.771	1	0.7714	1
table_b5	coffman_bilingual_republican, constant_se	0.032	0.032	1	0.03183	1
table_b5	coffman_monolingual_democrat, N	579	579	1	579	1
table_b5	coffman_monolingual_democrat, Z_ad	-0.141	-0.141	1	-0.1405	1
table_b5	coffman_monolingual_democrat, Z_ad_se	0.033	0.033	1	0.0334	1
table_b5	coffman_monolingual_democrat, constant	0.281	0.281	1	0.2814	1
table_b5	coffman_monolingual_democrat, constant_se	0.026	0.026	1	0.02622	1
table_b5	coffman_monolingual_republican, N	491	491	1	491	1
table_b5	coffman_monolingual_republican, Z_ad	-0.168	-0.168	1	-0.1678	1
table_b5	coffman_monolingual_republican, Z_ad_se	0.040	0.04	1	0.03999	1
table_b5	coffman_monolingual_republican, constant	0.816	0.816	1	0.8162	1
table_b5	coffman_monolingual_republican, constant_se	0.024	0.024	1	0.02353	1
table_b5	vela_bilingual_democrat, N	1029	1029	1	1029	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_b5	vela_bilingual_democrat, Z_ad	0.064	0.064	1	0.06365	1
table_b5	vela_bilingual_democrat, Z_ad_se	0.026	0.026	1	0.02592	1
table_b5	vela_bilingual_democrat, constant	0.744	0.744	1	0.7443	1
table_b5	vela_bilingual_democrat, constant_se	0.019	0.019	1	0.01908	1
table_b5	vela_bilingual_republican, N	335	335	1	335	1
table_b5	vela_bilingual_republican, Z_ad	-0.010	-0.01	1	-0.01023	1
table_b5	vela_bilingual_republican, Z_ad_se	0.050	0.05	1	0.04955	1
table_b5	vela_bilingual_republican, constant	0.292	0.292	1	0.2917	1
table_b5	vela_bilingual_republican, constant_se	0.035	0.035	1	0.03517	1
table_b5	vela_monolingual_democrat, N	579	579	1	579	1
table_b5	vela_monolingual_democrat, Z_ad	-0.044	-0.044	1	-0.04366	1
table_b5	vela_monolingual_democrat, Z_ad_se	0.040	0.04	1	0.04025	1
table_b5	vela_monolingual_democrat, constant	0.648	0.648	1	0.6477	1
table_b5	vela_monolingual_democrat, constant_se	0.029	0.029	1	0.02855	1
table_b5	vela_monolingual_republican, N	491	491	1	491	1
table_b5	vela_monolingual_republican, Z_ad	-0.011	-0.011	1	-0.01121	1
table_b5	vela_monolingual_republican, Z_ad_se	0.031	0.031	1	0.03063	1
table_b5	vela_monolingual_republican, constant	0.138	0.138	1	0.1378	1
table_b5	vela_monolingual_republican, constant_se	0.022	0.022	1	0.02167	1
table_b6	coffman_bilingual_democrat, N	1029	1029	1	1029	1
table_b6	coffman_bilingual_democrat, Z_ad	0.146	0.146	1	0.1461	1
table_b6	coffman_bilingual_democrat, Z_ad_se	0.102	0.102	1	0.1024	1
table_b6	coffman_bilingual_democrat, constant	4.722	4.722	1	4.722	1
table_b6	coffman_bilingual_democrat, constant_se	0.072	0.072	1	0.07227	1
table_b6	coffman_bilingual_republican, N	335	335	1	335	1
table_b6	coffman_bilingual_republican, Z_ad	-0.303	-0.303	1	-0.303	1
table_b6	coffman_bilingual_republican, Z_ad_se	0.159	0.159	1	0.1593	1
table_b6	coffman_bilingual_republican, constant	5.634	5.634	1	5.634	1
table_b6	coffman_bilingual_republican, constant_se	0.112	0.112	1	0.1115	1
table_b6	coffman_monolingual_democrat, N	579	579	1	579	1
table_b6	coffman_monolingual_democrat, Z_ad	-0.702	-0.702	1	-0.7019	1
table_b6	coffman_monolingual_democrat, Z_ad_se	0.114	0.114	1	0.1141	1
table_b6	coffman_monolingual_democrat, constant	4.969	4.969	1	4.969	1
table_b6	coffman_monolingual_democrat, constant_se	0.086	0.086	1	0.08582	1
table_b6	coffman_monolingual_republican, N	490	490	1	490	1
table_b6	coffman_monolingual_republican, Z_ad	-1.004	-1.004	1	-1.004	1
table_b6	coffman_monolingual_republican, Z_ad_se	0.123	0.123	1	0.1227	1
table_b6	coffman_monolingual_republican, constant	5.467	5.467	1	5.467	1
table_b6	coffman_monolingual_republican, constant_se	0.083	0.083	1	0.08332	1
table_b6	vela_bilingual_democrat, N	1028	1028	1	1028	1
table_b6	vela_bilingual_democrat, Z_ad	0.116	0.116	1	0.1157	1
table_b6	vela_bilingual_democrat, Z_ad_se	0.082	0.082	1	0.08197	1
table_b6	vela_bilingual_democrat, constant	5.317	5.317	1	5.317	1
table_b6	vela_bilingual_democrat, constant_se	0.057	0.057	1	0.05735	1
table_b6	vela_bilingual_republican, N	335	335	1	335	1
table_b6	vela_bilingual_republican, Z_ad	-0.247	-0.247	1	-0.2468	1
table_b6	vela_bilingual_republican, Z_ad_se	0.160	0.16	1	0.1599	1
table_b6	vela_bilingual_republican, constant	4.792	4.792	1	4.792	1
table_b6	vela_bilingual_republican, constant_se	0.109	0.109	1	0.1094	1
table_b6	vela_monolingual_democrat, N	578	578	1	578	1
table_b6	vela_monolingual_democrat, Z_ad	-0.608	-0.608	1	-0.6084	1
table_b6	vela_monolingual_democrat, Z_ad_se	0.102	0.102	1	0.1024	1
table_b6	vela_monolingual_democrat, constant	5.129	5.129	1	5.129	1
table_b6	vela_monolingual_democrat, constant_se	0.078	0.078	1	0.07768	1
table_b6	vela_monolingual_republican, N	491	491	1	491	1
table_b6	vela_monolingual_republican, Z_ad	-0.399	-0.399	1	-0.399	1
table_b6	vela_monolingual_republican, Z_ad_se	0.118	0.118	1	0.1184	1
table_b6	vela_monolingual_republican, constant	4.437	4.437	1	4.437	1
table_b6	vela_monolingual_republican, constant_se	0.081	0.081	1	0.08147	1
table_b7	coffman_bilingual_democrat, N	1029	1029	1	1029	1
table_b7	coffman_bilingual_democrat, Z_ad	0.078	0.078	1	0.07805	1
table_b7	coffman_bilingual_democrat, Z_ad_se	0.028	0.028	1	0.02844	1
table_b7	coffman_bilingual_democrat, constant	0.662	0.662	1	0.6623	1
table_b7	coffman_bilingual_democrat, constant_se	0.020	0.02	1	0.02045	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_b7	coffman_bilingual_republican, N	335	335	1	335	1
table_b7	coffman_bilingual_republican, Z_ad	-0.055	-0.055	1	-0.055	1
table_b7	coffman_bilingual_republican, Z_ad_se	0.039	0.039	1	0.03892	1
table_b7	coffman_bilingual_republican, constant	0.880	0.88	1	0.88	1
table_b7	coffman_bilingual_republican, constant_se	0.025	0.025	1	0.02464	1
table_b7	coffman_monolingual_democrat, N	575	575	1	575	1
table_b7	coffman_monolingual_democrat, Z_ad	-0.133	-0.133	1	-0.1326	1
table_b7	coffman_monolingual_democrat, Z_ad_se	0.037	0.037	1	0.03735	1
table_b7	coffman_monolingual_democrat, constant	0.782	0.782	1	0.7816	1
table_b7	coffman_monolingual_democrat, constant_se	0.024	0.024	1	0.02418	1
table_b7	coffman_monolingual_republican, N	490	490	1	490	1
table_b7	coffman_monolingual_republican, Z_ad	-0.181	-0.181	1	-0.1814	1
table_b7	coffman_monolingual_republican, Z_ad_se	0.037	0.037	1	0.03696	1
table_b7	coffman_monolingual_republican, constant	0.879	0.879	1	0.8787	1
table_b7	coffman_monolingual_republican, constant_se	0.020	0.02	1	0.01983	1
table_b7	vela_bilingual_democrat, N	1029	1029	1	1029	1
table_b7	vela_bilingual_democrat, Z_ad	0.040	0.04	1	0.03962	1
table_b7	vela_bilingual_democrat, Z_ad_se	0.017	0.017	1	0.01734	1
table_b7	vela_bilingual_democrat, constant	0.895	0.895	1	0.895	1
table_b7	vela_bilingual_democrat, constant_se	0.013	0.013	1	0.0134	1
table_b7	vela_bilingual_republican, N	335	335	1	335	1
table_b7	vela_bilingual_republican, Z_ad	-0.049	-0.049	1	-0.04926	1
table_b7	vela_bilingual_republican, Z_ad_se	0.047	0.047	1	0.04744	1
table_b7	vela_bilingual_republican, constant	0.774	0.774	1	0.7738	1
table_b7	vela_bilingual_republican, constant_se	0.032	0.032	1	0.03237	1
table_b7	vela_monolingual_democrat, N	576	576	1	576	1
table_b7	vela_monolingual_democrat, Z_ad	-0.113	-0.113	1	-0.1126	1
table_b7	vela_monolingual_democrat, Z_ad_se	0.032	0.032	1	0.03195	1
table_b7	vela_monolingual_democrat, constant	0.872	0.872	1	0.8719	1
table_b7	vela_monolingual_democrat, constant_se	0.020	0.02	1	0.01997	1
table_b7	vela_monolingual_republican, N	490	490	1	490	1
table_b7	vela_monolingual_republican, Z_ad	-0.226	-0.226	1	-0.2256	1
table_b7	vela_monolingual_republican, Z_ad_se	0.043	0.043	1	0.04334	1
table_b7	vela_monolingual_republican, constant	0.709	0.709	1	0.7087	1
table_b7	vela_monolingual_republican, constant_se	0.029	0.029	1	0.02857	1
table_b8	coffman_bilingual_democrat, N	1028	1028	1	1028	1
table_b8	coffman_bilingual_democrat, Z_ad	0.128	0.128	1	0.1284	1
table_b8	coffman_bilingual_democrat, Z_ad_se	0.052	0.052	1	0.05224	1
table_b8	coffman_bilingual_democrat, constant	1.650	1.65	1	1.65	1
table_b8	coffman_bilingual_democrat, constant_se	0.036	0.036	1	0.03626	1
table_b8	coffman_bilingual_republican, N	335	335	1	335	1
table_b8	coffman_bilingual_republican, Z_ad	-0.139	-0.139	1	-0.1391	1
table_b8	coffman_bilingual_republican, Z_ad_se	0.084	0.084	1	0.08357	1
table_b8	coffman_bilingual_republican, constant	2.183	2.183	1	2.183	1
table_b8	coffman_bilingual_republican, constant_se	0.051	0.051	1	0.0513	1
table_b8	coffman_monolingual_democrat, N	577	577	1	577	1
table_b8	coffman_monolingual_democrat, Z_ad	-0.179	-0.179	1	-0.1786	1
table_b8	coffman_monolingual_democrat, Z_ad_se	0.067	0.067	1	0.06671	1
table_b8	coffman_monolingual_democrat, constant	1.895	1.895	1	1.895	1
table_b8	coffman_monolingual_democrat, constant_se	0.047	0.047	1	0.04677	1
table_b8	coffman_monolingual_republican, N	489	489	1	489	1
table_b8	coffman_monolingual_republican, Z_ad	-0.491	-0.491	1	-0.491	1
table_b8	coffman_monolingual_republican, Z_ad_se	0.073	0.073	1	0.07269	1
table_b8	coffman_monolingual_republican, constant	2.099	2.099	1	2.099	1
table_b8	coffman_monolingual_republican, constant_se	0.043	0.043	1	0.04332	1
table_b8	vela_bilingual_democrat, N	1029	1029	1	1029	1
table_b8	vela_bilingual_democrat, Z_ad	0.110	0.11	1	0.1098	1
table_b8	vela_bilingual_democrat, Z_ad_se	0.042	0.042	1	0.04218	1
table_b8	vela_bilingual_democrat, constant	2.076	2.076	1	2.076	1
table_b8	vela_bilingual_democrat, constant_se	0.029	0.029	1	0.02915	1
table_b8	vela_bilingual_republican, N	335	335	1	335	1
table_b8	vela_bilingual_republican, Z_ad	-0.073	-0.073	1	-0.07339	1
table_b8	vela_bilingual_republican, Z_ad_se	0.096	0.096	1	0.09595	1
table_b8	vela_bilingual_republican, constant	1.744	1.744	1	1.744	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_b8	vela_bilingual_republican, constant_se	0.071	0.071	1	0.07064	1
table_b8	vela_monolingual_democrat, N	577	577	1	577	1
table_b8	vela_monolingual_democrat, Z_ad	-0.181	-0.181	1	-0.1807	1
table_b8	vela_monolingual_democrat, Z_ad_se	0.062	0.062	1	0.06249	1
table_b8	vela_monolingual_democrat, constant	2.032	2.032	1	2.032	1
table_b8	vela_monolingual_democrat, constant_se	0.041	0.041	1	0.04138	1
table_b8	vela_monolingual_republican, N	490	490	1	490	1
table_b8	vela_monolingual_republican, Z_ad	-0.192	-0.192	1	-0.1922	1
table_b8	vela_monolingual_republican, Z_ad_se	0.079	0.079	1	0.07948	1
table_b8	vela_monolingual_republican, constant	1.480	1.48	1	1.48	1
table_b8	vela_monolingual_republican, constant_se	0.052	0.052	1	0.05165	1
table_c9	bush_bilingual, N	1849	1849	1	1849	1
table_c9	bush_bilingual, Z_ad	0.196	0.196	1	0.1965	1
table_c9	bush_bilingual, Z_ad_se	0.093	0.093	1	0.09326	1
table_c9	bush_bilingual, Z_survey	0.022	0.022	1	0.02176	1
table_c9	bush_bilingual, Z_survey_se	0.093	0.093	1	0.09326	1
table_c9	bush_bilingual, aic	2560.860	2561	1	2561	1
table_c9	bush_bilingual, constant	-0.203	-0.203	1	-0.2028	1
table_c9	bush_bilingual, constant_se	0.081	0.081	1	0.08104	1
table_c9	bush_bilingual, log_likelihood	-1277.430	-1277	1	-1277	1
table_c9	coffman_bilingual, N	1681	1681	1	1681	1
table_c9	coffman_bilingual, Z_ad	0.012	0.012	1	0.01158	1
table_c9	coffman_bilingual, Z_ad_se	0.101	0.101	1	0.101	1
table_c9	coffman_bilingual, Z_survey	-0.119	-0.119	1	-0.119	1
table_c9	coffman_bilingual, Z_survey_se	0.101	0.101	1	0.1011	1
table_c9	coffman_bilingual, aic	2224.255	2224	1	2224	1
table_c9	coffman_bilingual, constant	-0.471	-0.471	1	-0.4712	1
table_c9	coffman_bilingual, constant_se	0.085	0.085	1	0.08475	1
table_c9	coffman_bilingual, log_likelihood	-1109.128	-1109	1	-1109	1
table_c9	coffman_monolingual, N	1343	1343	1	1343	1
table_c9	coffman_monolingual, Z_ad	-0.780	-0.78	1	-0.7796	1
table_c9	coffman_monolingual, Z_ad_se	0.113	0.113	1	0.1133	1
table_c9	coffman_monolingual, aic	1786.530	1787	1	1787	1
table_c9	coffman_monolingual, constant	0.051	0.051	1	0.05072	1
table_c9	coffman_monolingual, constant_se	0.075	0.075	1	0.07508	1
table_c9	coffman_monolingual, log_likelihood	-891.265	-891.3	1	-891.3	1
table_c9	vela_bilingual, N	1680	1680	1	1680	1
table_c9	vela_bilingual, Z_ad	0.203	0.203	1	0.2026	1
table_c9	vela_bilingual, Z_ad_se	0.100	0.1	1	0.09986	1
table_c9	vela_bilingual, Z_survey	0.304	0.304	1	0.3042	1
table_c9	vela_bilingual, Z_survey_se	0.100	0.1	1	0.1001	1
table_c9	vela_bilingual, aic	2262.869	2263	1	2263	1
table_c9	vela_bilingual, constant	0.137	0.137	1	0.1369	1
table_c9	vela_bilingual, constant_se	0.084	0.084	1	0.08419	1
table_c9	vela_bilingual, log_likelihood	-1128.434	-1128	1	-1128	1
table_c9	vela_monolingual, N	1343	1343	1	1343	1
table_c9	vela_monolingual, Z_ad	-0.088	-0.088	1	-0.08778	1
table_c9	vela_monolingual, Z_ad_se	0.114	0.114	1	0.114	1
table_c9	vela_monolingual, aic	1752.085	1752	1	1752	1
table_c9	vela_monolingual, constant	-0.550	-0.55	1	-0.5497	1
table_c9	vela_monolingual, constant_se	0.080	0.08	1	0.07991	1
table_c9	vela_monolingual, log_likelihood	-874.043	-874	1	-874	1
table_c10	bush_bilingual, N	1858	1858	1	1858	1
table_c10	bush_bilingual, Z_ad	0.193	0.193	1	0.1928	1
table_c10	bush_bilingual, Z_ad_se	0.106	0.106	1	0.1055	1
table_c10	bush_bilingual, Z_survey	-0.179	-0.179	1	-0.1793	1
table_c10	bush_bilingual, Z_survey_se	0.105	0.105	1	0.1055	1
table_c10	bush_bilingual, aic	2143.540	2144	1	2144	1
table_c10	bush_bilingual, constant	1.020	1.02	1	1.02	1
table_c10	bush_bilingual, constant_se	0.091	0.091	1	0.09133	1
table_c10	bush_bilingual, log_likelihood	-1068.770	-1069	1	-1069	1
table_c10	coffman_bilingual, N	1680	1680	1	1680	1
table_c10	coffman_bilingual, Z_ad	0.180	0.18	1	0.1803	1
table_c10	coffman_bilingual, Z_ad_se	0.112	0.112	1	0.1118	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
table_c10	coffman_bilingual, Z_survey	-0.233	-0.233	1	-0.2331	1
table_c10	coffman_bilingual, Z_survey_se	0.111	0.111	1	0.1114	1
table_c10	coffman_bilingual, aic	1926.946	1927	1	1927	1
table_c10	coffman_bilingual, constant	1.072	1.072	1	1.072	1
table_c10	coffman_bilingual, constant_se	0.094	0.094	1	0.0943	1
table_c10	coffman_bilingual, log_likelihood	-960.473	-960.5	1	-960.5	1
table_c10	coffman_monolingual, N	1337	1337	1	1337	1
table_c10	coffman_monolingual, Z_ad	-0.820	-0.82	1	-0.8203	1
table_c10	coffman_monolingual, Z_ad_se	0.128	0.128	1	0.1283	1
table_c10	coffman_monolingual, aic	1488.777	1489	1	1489	1
table_c10	coffman_monolingual, constant	1.483	1.483	1	1.483	1
table_c10	coffman_monolingual, constant_se	0.097	0.097	1	0.09678	1
table_c10	coffman_monolingual, log_likelihood	-742.389	-742.4	1	-742.4	1
table_c10	vela_bilingual, N	1680	1680	1	1680	1
table_c10	vela_bilingual, Z_ad	0.148	0.148	1	0.1485	1
table_c10	vela_bilingual, Z_ad_se	0.142	0.142	1	0.142	1
table_c10	vela_bilingual, Z_survey	-0.201	-0.201	1	-0.2015	1
table_c10	vela_bilingual, Z_survey_se	0.142	0.142	1	0.1417	1
table_c10	vela_bilingual, aic	1351.811	1352	1	1352	1
table_c10	vela_bilingual, constant	1.860	1.86	1	1.86	1
table_c10	vela_bilingual, constant_se	0.122	0.122	1	0.1219	1
table_c10	vela_bilingual, log_likelihood	-672.905	-672.9	1	-672.9	1
table_c10	vela_monolingual, N	1336	1336	1	1336	1
table_c10	vela_monolingual, Z_ad	-0.743	-0.743	1	-0.7427	1
table_c10	vela_monolingual, Z_ad_se	0.123	0.123	1	0.123	1
table_c10	vela_monolingual, aic	1587.141	1587	1	1587	1
table_c10	vela_monolingual, constant	1.268	1.268	1	1.268	1
table_c10	vela_monolingual, constant_se	0.093	0.093	1	0.09305	1
table_c10	vela_monolingual, log_likelihood	-791.570	-791.6	1	-791.6	1
text	4 is said to indicate a lot	4			3	0
text	4 is said to indicate greater confidence	4			3	0
text	A 2 x 2 design: advertisement-language arms	2			2	1
text	A 2 x 2 design: survey-language arms	2			2	1
text	A Spanish strategy can lose ground above a half-bilingual electorate					
text	Among monolinguals the pattern does not differ by party					
text	Appendix A covers all four dependent variables	4			4	1
text	Average marginal effects match OLS to the second decimal place					
text	Balance test: chi-squared statistic	7.3	7.258	1	7.258	1
text	Balance test: degrees of freedom	7	7	1	7	1
text	Balance test: p-value	0.40	0.4025	1	0.4025	1
text	Bilingual Democrats respond positively and bilingual Republicans negatively					
text	Bilingual effects on candidate caring are on the order of 2 to 3 points					
text	Both bilingual advertisement effects are significant					
text	Both linked fate estimates are significant					
text	Bush and Vela bilinguals were about five points more likely to support	5	4.9	1	4.876	1
text	Bush bilingual advertisement effect	4.9	4.9	1	4.897	1
text	Bush bilingual advertisement effect, standard error	2.3	2.3	1	2.322	1
text	Bush experiment: bilinguals recruited	2866			2866	1
text	Bush experiment: bilinguals who passed the quiz	1862			1862	1
text	Candidate cares is coded 0 for does not care	0			0	1
text	Candidate cares is coded 1 for cares	1			1	1
text	Candidate preference is coded 0 otherwise	0			0	1
text	Candidate preference is coded 1 for the advertising candidate	1			1	1
text	Coffman monolingual Democrat advertisement effect	-14.1	-14.1	1	-14.05	1
text	Coffman monolingual Democrat effect, standard error	3.3	3.3	1	3.34	1
text	Coffman monolingual Republican advertisement effect	-16.8	-16.8	1	-16.78	1
text	Coffman monolingual Republican effect, standard error	4.0	4	1	3.999	1
text	Coffman monolingual advertisement effect	-18.7	-18.7	1	-18.72	1
text	Coffman monolingual advertisement effect, standard error	2.6	2.6	1	2.645	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
text	Coffman monolingual effect on candidate caring	-15.5	-15.5	1	-15.52	1
text	Coffman monolingual effect on candidate caring, standard error	2.4	2.4	1	2.389	1
text	Coffman's support in the monolingual control group	51	51.3	1	51.27	1
text	Confidence is said to run from 1	1			0	0
text	Confidence is said to run to 4	4			3	0
text	Effect of the Spanish survey on linked fate, Bush experiment	0.24	0.243	1	0.2427	1
text	Effect of the Spanish survey on support for Vela	7.3	7.3	1	7.296	1
text	Effect of the Spanish survey on support for Vela, standard error	2.4	2.4	1	2.39	1
text	Effect on liking Bush	0.167	0.167	1	0.1668	1
text	Effect on liking Bush, standard error	0.075	0.075	1	0.07533	1
text	Effect on linked fate, Bush experiment, standard error	0.04	0.04	1	0.0403	1
text	Effect on linked fate, Vela and Coffman experiments	0.13	0.131	1	0.1311	1
text	Effect on linked fate, Vela and Coffman, standard error	0.04	0.041	1	0.04127	1
text	Experiment 1's bilingual sample	1862			1862	1
text	Experiments 2 and 3: bilingual sample	1681			1681	1
text	Experiments 2 and 3: monolingual sample	1344			1344	1
text	Five outcome measures	5			5	1
text	Formal tests of non-response against treatment assignment					
text	Half the subjects took the survey in English					
text	Interactions significant at the 10 per cent level	2			2	1
text	Interactions significant at the 5 per cent level	0			0	1
text	Item non-response moves the analysed sample only slightly					
text	Liking is measured on a seven-point scale	7			7	1
text	Liking runs from 1	1			1	1
text	Liking runs to 7	7			7	1
text	Linked fate is said to run from 1	1			0	0
text	Linked fate is said to run to 4	4			3	0
text	Preview: Spanish ad raises Bush support by about five points	5	4.9	1	4.897	1
text	Sampling: Bush respondents who passed the language quiz	1862			1862	1
text	Sampling: Latinos who responded to the Bush survey	2866			2866	1
text	Sampling: bilinguals who responded to the Vela and Coffman survey	2233			2233	1
text	Sampling: nationally representative subjects supplied	2230			2230	1
text	Sampling: the same count, restated	1862			1862	1
text	Sampling: those who did not pass the language quiz	1344			1344	1
text	Sampling: those who passed the language quiz	1681			1681	1
text	Six advertisements in all	6			6	1
text	Spanish ad raises bilingual support by five percentage points	5	4.9	1	4.876	1
text	Table 2 row label: Education(5 levels)	5			5	1
text	Table 2 row label: Income(7 levels)	7			7	1
text	Table 2 row label: Income(9 levels)	9			9	1
text	Table 5 covers all three experiments	3			3	1
text	The Republican and Democrat effects do not differ significantly					
text	The Spanish-language survey is said to be in supplemental Appendix C					
text	The Vela monolingual effect cannot be distinguished from zero					
text	The Vela point estimate is identical to the Bush one					
text	The calibration supposes a five-point bilingual effect	5			5	1
text	The calibration supposes a negative fifteen-point monolingual effect	-15			-15	1
text	The interaction coefficients are small and insignificant					
text	The language quiz is said to be in supplemental Appendix D					
text	The logit tables are said to be in supplemental Appendix A					
text	The simulated electorate runs from 0 per cent bilingual	0			0	1

Table 5: Ground truth: the published value against the value the deposited scripts print and the value the maintained rewrite computes. A blank Archive column means no deposited script prints the quantity. *(continued)*

Location	Claim	Paper	Archive	Match	Rewrite	Match
text	The simulated electorate runs to 100 per cent bilingual	100			100	1
text	Three candidates produced matched advertisements	3			3	1
text	Three randomized survey experiments	3			3	1
text	Three separate experiments were conducted	3			3	1
text	Twelve interaction terms	12			12	1
text	Twelve interaction terms, restated	12			12	1
text	Two dependent variables are binary	2			2	1
text	Two of three experiments raise bilingual support by about five points					
text	Vela and Coffman experiments: bilinguals recruited	2233			2233	1
text	Vela and Coffman experiments: bilinguals who passed the quiz	1681			1681	1
text	Vela bilingual advertisement effect	4.9	4.9	1	4.855	1
text	Vela bilingual advertisement effect, standard error	2.4	2.4	1	2.39	1
text	Vela monolingual advertisement effect	-2	-2	1	-2.012	1
text	Vela monolingual effect on candidate caring	-15.2	-15.2	1	-15.2	1
text	Vela monolingual effect on candidate caring, standard error	2.5	2.5	1	2.466	1

5 Table 2 has no code

Table 2 compares the bilingual Lucid sample in the Bush experiment against bilingual respondents in the 2006 Latino National Survey and the 2012 Pew National Survey of Latinos. No script in the deposit computes any part of it. The archive's `FC_bilinguals_in_text_stats.R` ends with a summary of five covariates from `study_3.csv`, which is a different sample and a different set of variables; nothing produces Table 2.

The rewrite adds `table_2_sample_comparison.R`, and all 47 values reproduce. The Lucid column's 17 come from the deposited `study_1.csv`, and getting them requires one undocumented recode: `educ_5` stores "Not Found" as 99 for 96 of the 1,862 respondents, and leaving those in returns a mean of 7.95 against the article's 3.01. The 30 in the LNS and Pew columns are weighted means among the bilingual respondents of the two national surveys, carrying the standard error of a weighted mean rather than of an unweighted one.

Those two surveys are licensed and neither may be redistributed, so neither the deposit nor this repository carries them. What this repository carries is the aggregate, in `maintained/output/table_2_sample_comparison.csv`, which is what the published table prints and the article therefore already made public. The script recomputes the two columns from the microdata when they are present at `licensed_data/national_surveys_stacked.rdata`, and without them leaves the committed values standing and says why.

Table 6: Table 2: the article's value against the rewrite, which is the only code that computes it.

Column	Quantity	Paper	Rewrite mean	Rewrite SE	Non-missing

lucid	female	0.70	0.70	0.01	1862
lucid	age	34.80	34.80	0.30	1862
lucid	education_5	3.01	3.01	0.03	1766
lucid	mexican	0.49	0.49	0.01	1862
lucid	cuban	0.07	0.07	0.01	1862
lucid	other_hispanic	0.44	0.44	0.01	1862
lucid	income_7	4.11	4.11	0.05	1713
lucid	income_9	4.49	4.49	0.05	1713
lucid	n	1862	1862.00	NA	1862
lns	female	0.52	0.52	0.01	4184
lns	age	33.78	33.78	0.29	4036
lns	education_5	2.49	2.49	0.02	4184
lns	mexican	0.67	0.67	0.01	4184
lns	cuban	0.04	0.04	0.00	4184
lns	other_hispanic	0.29	0.29	0.01	4184
lns	income_7	4.06	4.06	0.04	3478
lns	n	4184	4184.00	NA	4184
pew	female	0.46	0.46	0.02	715
pew	age	40.11	40.11	0.84	657
pew	education_5	2.45	2.45	0.05	706
pew	mexican	0.59	0.59	0.02	715
pew	cuban	0.05	0.05	0.01	715
pew	other_hispanic	0.36	0.36	0.02	715
pew	income_9	4.09	4.09	0.11	640
pew	n	715	715.00	NA	715

6 Maintained rewrite

The rewrite lives in `maintained/`: a shared `helpers.R` and sixteen scripts covering the seven main-text tables, the ten appendix tables, the three figures, the numbers stated in prose, the design quantities the article states without tabling them, and the second instrument. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used.

6.1 Architecture

`helpers.R` loads the three deposited CSVs and builds the three analysis subsamples, which is the one piece of data preparation the archive repeats inside each of its eight scripts. Each table script then declares its models as a list of specifications and fits them in one pass, so the outcome, the treatment indicator and the sample of every column are visible in a single block.

Standard errors are the main substitution. The archive fits with `lm()` and passes `estimatr::starprep()` into `stargazer` to get HC2 standard errors, which means the coefficients and the standard errors printed beside them come from two different objects. The rewrite fits with `lm_robust(se_type = "HC2")`, which is the same estimator and the same standard errors in one call, and writes a tidy CSV rather than a LaTeX table, so the table and any downstream check cannot disagree.

`text_in_text_claims.R` reads the committed table output and reports the numbers the article states in prose. It reads no data of its own, so an in-text claim and the table it is drawn from cannot disagree.

6.2 Deprecated patterns replaced

Table 7: Patterns replaced in the maintained rewrite, and whether the replacement was forced or chosen.

Original pattern	Replacement	Required?
‘rm(list = ls())‘	(omitted)	no
commented ‘setwd()‘ instruction	‘here::here()‘	no
‘lm()‘ + ‘starprep()‘ into ‘stargazer‘	‘estimatr::lm_robust(se_type = ”HC2”)‘	no
‘stargazer‘ LaTeX to the console	‘write_csv()‘ to ‘output/‘	no
‘geom_errorbarh()‘	‘geom_linerange()‘	yes, deprecated in ggplot2 4.0.0
‘ggstance::position_dodgev()‘	‘position_dodge()‘	no, ggstance still installs
‘margins::margins()‘	‘marginaleffects::avg_slopes()‘	no, margins 0.3.28 released July 2024
‘ggsave()‘ to a bare relative path	‘ggsave(here::here(”maintained”, ”output”, ...))‘	yes, it overwrites the deposit
‘expand.grid()‘ + ‘for‘ loop over rows	‘expand_grid()‘ + ‘mutate()‘	no
‘within()‘ blocks over base indexing	‘mutate()‘ paragraphs	no
magrittr pipe	native pipe	no

Only two of the eleven substitutions were forced. `geom_errorbarh()` is deprecated and the horizontal intervals now come from `geom_linerange()`, which the project's style guide prefers in any case. The `ggsave()` paths had to move because the originals write into the deposit. Everything else is a choice, and the table says so rather than implying a package failure that did not occur. `margins` in particular is alive, and running it beside `avg_slopes()` on the same fitted models returns the same average marginal effects to within 1e-4, the gap being numerical differentiation in one and not the other.

7 Figures

Figure 1 and Figure 2 are the two the deposit pre-renders, and the rewrite reproduces both. The estimates behind every point are written to `figure_1_main_effects.csv` and `figure_2_het_fx_party.csv`, so the figures are diffable rather than only viewable. Neither deposited figure script prints or saves a single number, so those two CSVs are the only form in which the plotted estimates exist anywhere; `build_ground_truth.R` compares all 52 of them, on estimate, standard error and both confidence limits, against the models the deposited scripts specify.

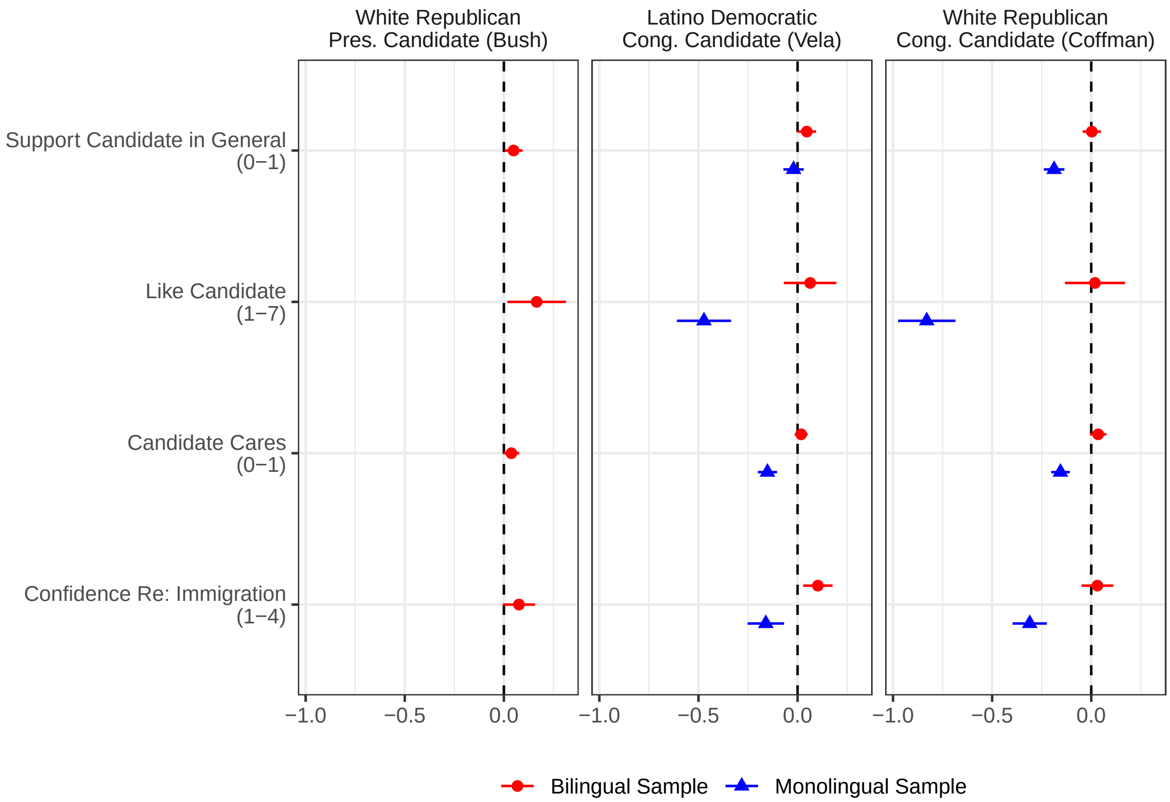

Figure 1: Figure 1 as reproduced by the maintained rewrite.

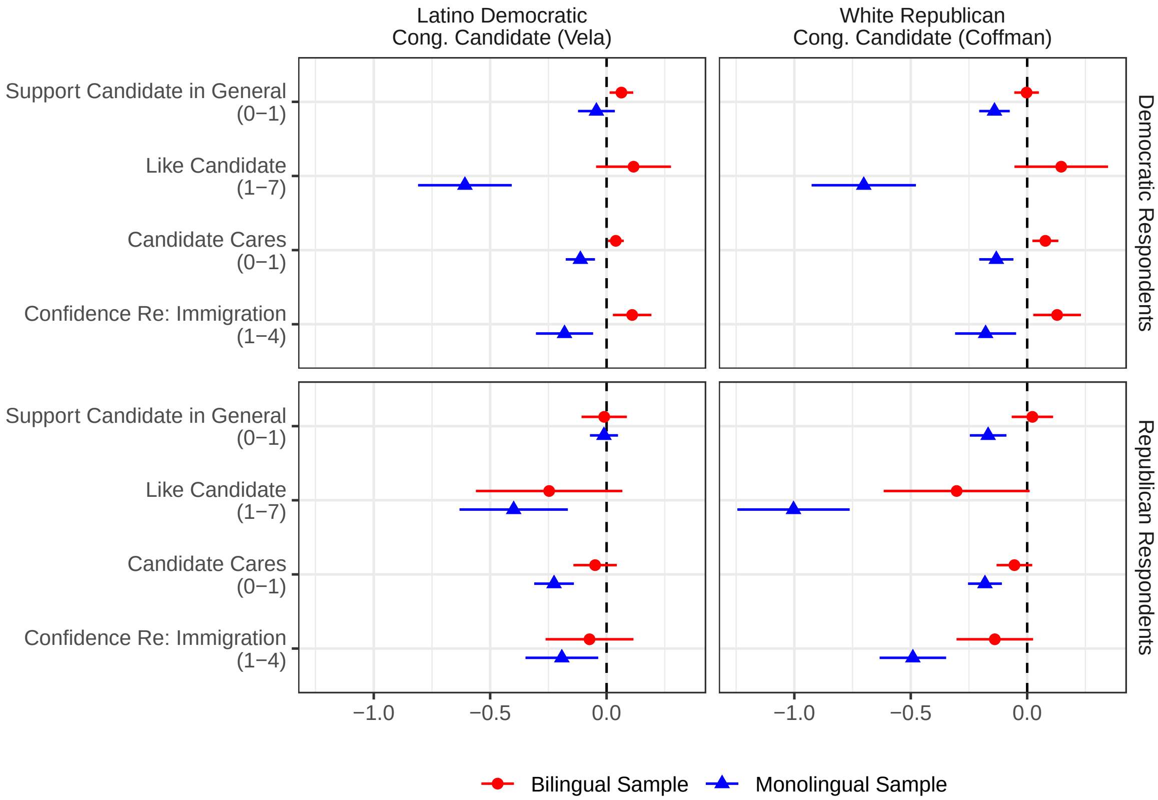

Figure 2: Figure 2 as reproduced by the maintained rewrite.

Figure 3 is the exception to the rule that every number in `output/` comes from data, and it is worth being precise about why. The figure is not an estimate but a calibration exercise: it maps the net electoral payoff of a Spanish-language strategy over two quantities the experiments do not measure, the share of bilinguals in the electorate and the probability that a monolingual sees the ad anyway. The two effect sizes it holds fixed, positive 5 points among bilinguals and negative 15 among monolinguals, are stated in the article's own prose as a supposition. Substituting the estimated effects would draw a different figure than the paper drew, which is a reanalysis rather than a maintenance fix, so the assumption stays and `text_in_text_claims.R` reports the estimates it approximates alongside it. `figure_3_simulation.csv` records the surface on an eleven-by-eleven grid and the break-even mistargeting risk at each bilingual share, which is what the figure is read for. The coarse grid is computed on its own rather than filtered out of the fine one the figure draws: `seq(0, 1, by = 0.01)` and `seq(0, 1, by = 0.1)` do not agree on 0.3 or 0.6 in floating point, so a membership filter silently drops two of the eleven lines on each axis.

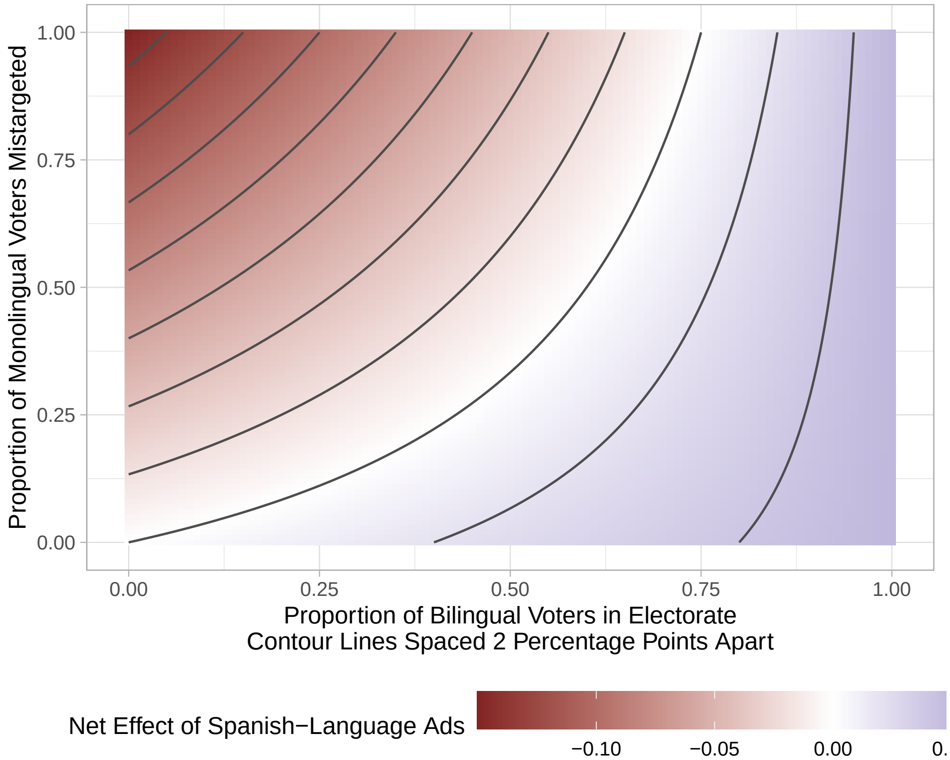

Figure 3: Figure 3 as reproduced by the maintained rewrite.

8 The extraction and the two instruments

The comparison above is float-shaped: it asks whether the numbers in the article's tables reproduce. Most of what a reader would quote is not in a table, so the repository carries a second, sentence-shaped instrument beside it.

`ground_truth/published_claims.csv` is the extraction. The article and the appendix were read line by line and every numeric token recorded with its location, its type and, where it has one, the string the page prints. It holds 204 rows: 59 `pipeline` claims (a quantity the analysis produces), 18 `descriptive` claims (a statement about shape, sign or count with a truth value rather than a number), 73 `definitional` (scale endpoints, field dates, category counts), 51 `structural` (counts of the paper's own sections, floats and cross-references) and 3 `transcribed` (values quoted from other papers, which cannot drift). The coverage boundary is stated plainly: the extraction covers every numeric token in the article's prose, notes, captions, table titles and row labels, and in the appendix's prose and its contents page, together with one row per published float recording how many numbers that float carries. It does **not** repeat the cells of the typeset tables, which are transcribed one by one in `published_values.csv` and compared in the ground truth. Excluded, and named here so the exclusions are auditable: the journal's own front matter and correspondence address, citation years and page or footnote locators inside citations, the reference list, running heads and page numbers, the ORCID, footnote markers, and the digits inside YouTube identifiers.

`maintained/in_text_claims.R` is the second instrument. It recomputes every quantity a sentence states, from `maintained/output/`, by a path of its own, and prints it beside the sentence. It never reads the ground truth, because agreeing with the comparison would prove nothing, and no published number is typed into it. Each line reads `CLAIM <id> = <value> || <label>`.

The coverage gate is the last step of `build_ground_truth.R`. It runs `in_text_claims.R` as a program in its own environment, captures what it prints, and requires that the 102 extraction rows needing a block are exactly the claims printed, in both directions; that the value each instrument reaches agrees with the other at the article's own precision; that every stored published value survives a round trip through its own recorded precision, checked before anything consumes it; and that the extraction and the ground truth, which are two hand transcriptions of the same pages, say the same thing. A block that errors, or prints nothing, fails the gate where a search for comment markers would pass it.

18 of the claims are descriptive and carry a computed truth value rather than a number: 13 hold, 3 do not, and 2 carry no verdict by design, one because the sentence hedges its magnitude and one because the deposit never computed the quantity. The 3 that do not hold are the appendix cross-references.

Table 8: Coverage per published float.

Float	Numbers it carries	Covered	Reproduce	Basis
table_1	8	0	0	no comparable number
table_2	47	47	47	published cells
table_3	16	16	16	published cells
table_4	9	0	0	no comparable number
table_5	31	31	31	published cells
table_6	31	31	31	published cells
table_7	31	31	31	published cells
table_8	31	31	31	published cells
table_9	10	10	10	published cells
figure_1	20	20	20	plotted estimates, compared against the deposit
figure_2	32	32	32	plotted estimates, compared against the deposit
figure_3	0	0	0	no comparable number
table_a1	27	27	27	published cells
table_a2	27	27	27	published cells
table_a3	27	27	27	published cells
table_a4	27	27	27	published cells
table_b5	40	40	40	published cells
table_b6	40	40	40	published cells
table_b7	40	40	40	published cells
table_b8	40	40	40	published cells
table_c9	41	41	41	published cells
table_c10	41	41	41	published cells

599 of the 616 numbers the article's floats carry are covered. The 17 that are not fall into two groups: Table 1's eight language indicators, which are notation, and Table 4's nine advertisement running times and election years, which the deposit does not record. Table 4 carries no estimate at all: it is the six advertisements' titles, running times, links and full transcripts in English and Spanish, which `ground_truth/extract_published_table_4.R` parses word by word out of the published table into `published_table_4_transcripts.csv`, so the one float in the article whose content is text rather than numbers is recorded rather than merely counted. Figures 1 and 2 print no estimate on their faces, so what each asserts is the number of estimates it plots, counted off the published page and recomputed from the figure's own CSV.

9 Errata

NA errors in the article are corrected in `flores_coppock_2018_errata.pdf`, rendered from `errata.qmd` at the root of this repository with every corrected value computed at render time rather than typed. Neither changes an estimate or a conclusion. They are numbered here as the note numbers them.

Entry 1 is that the Outcome Measures section gives the wrong endpoints for two of its five scales. Confidence in Candidate and Linked Fate are described as running from 1 to 4; the variables the article analyses run from 0 to 3, which is the scale the control means in Tables 8 and 9 are on. A shift of a scale leaves every treatment effect unchanged, so what the correction changes is how a reader should read the constants.

Entry 2 is that three of the article's five references to a lettered appendix section name the wrong section: the language quiz is said to be in Appendix D and is in Appendix E, the Spanish-language survey is said to be in Appendix C and is in Appendix D, and the logistic regression tables are said to be in Appendix A and are in Appendix C.

One descriptive claim sits close to the line and is recorded as holding. The article says that "among monolinguals, the pattern of treatment effects does not differ by respondent partisanship", glossing that in the next sentence as "Monolingual Republicans and Democrats alike respond negatively to the Spanish-language advertisement". All sixteen monolingual estimates are indeed negative. Two of the eight Democrat-Republican differences do separate from zero at conventional levels, so the sentence holds as the article itself defines it and not on a stricter reading.

10 In-text claims

The article states 78 numbers in a sentence rather than in a table, and makes a further 18 claims about shape, sign or count that carry a truth value instead of a number. The table below is the ground truth's own view of them; `in_text_claims.R` prints the same set independently, with a label naming the derivation behind each.

Table 9: Quantities the article states in a sentence, against the maintained rewrite.

Claim	Paper	Rewrite	Holds	Locus
4 is said to indicate a lot	4	3	0	paper_internal
4 is said to indicate greater confidence	4	3	0	paper_internal
A 2 x 2 design: advertisement-language arms	2	2	1	
A 2 x 2 design: survey-language arms	2	2	1	
A Spanish strategy can lose ground above a half-bilingual electorate			TRUE	
Among monolinguals the pattern does not differ by party			TRUE	
Appendix A covers all four dependent variables	4	4	1	
Average marginal effects match OLS to the second decimal place			TRUE	
Balance test: chi-squared statistic	7.3	7.258	1	
Balance test: degrees of freedom	7	7	1	
Balance test: p-value	0.40	0.4025	1	
Bilingual Democrats respond positively and bilingual Republicans negatively			TRUE	
Bilingual effects on candidate caring are on the order of 2 to 3 points				
Both bilingual advertisement effects are significant			TRUE	
Both linked fate estimates are significant			TRUE	
Bush and Vela bilinguals were about five points more likely to support	5	4.876	1	
Bush bilingual advertisement effect	4.9	4.897	1	
Bush bilingual advertisement effect, standard error	2.3	2.322	1	
Bush experiment: bilinguals recruited	2866	2866	1	
Bush experiment: bilinguals who passed the quiz	1862	1862	1	
Candidate cares is coded 0 for does not care	0	0	1	
Candidate cares is coded 1 for cares	1	1	1	
Candidate preference is coded 0 otherwise	0	0	1	
Candidate preference is coded 1 for the advertising candidate	1	1	1	
Coffman monolingual Democrat advertisement effect	-14.1	-14.05	1	
Coffman monolingual Democrat effect, standard error	3.3	3.34	1	
Coffman monolingual Republican advertisement effect	-16.8	-16.78	1	
Coffman monolingual Republican effect, standard error	4.0	3.999	1	
Coffman monolingual advertisement effect	-18.7	-18.72	1	
Coffman monolingual advertisement effect, standard error	2.6	2.645	1	
Coffman monolingual effect on candidate caring	-15.5	-15.52	1	
Coffman monolingual effect on candidate caring, standard error	2.4	2.389	1	
Coffman's support in the monolingual control group	51	51.27	1	
Confidence is said to run from 1	1	0	0	paper_internal
Confidence is said to run to 4	4	3	0	paper_internal
Effect of the Spanish survey on linked fate, Bush experiment	0.24	0.2427	1	
Effect of the Spanish survey on support for Vela	7.3	7.296	1	
Effect of the Spanish survey on support for Vela, standard error	2.4	2.39	1	
Effect on liking Bush	0.167	0.1668	1	
Effect on liking Bush, standard error	0.075	0.07533	1	
Effect on linked fate, Bush experiment, standard error	0.04	0.0403	1	
Effect on linked fate, Vela and Coffman experiments	0.13	0.1311	1	

Table 9: Quantities the article states in a sentence, against the maintained rewrite. *(continued)*

Claim	Paper	Rewrite	Holds	Locus
Effect on linked fate, Vela and Coffman, standard error	0.04	0.04127	1	
Experiment 1's bilingual sample	1862	1862	1	
Experiments 2 and 3: bilingual sample	1681	1681	1	
Experiments 2 and 3: monolingual sample	1344	1344	1	
Five outcome measures	5	5	1	
Formal tests of non-response against treatment assignment				archive
Half the subjects took the survey in English			TRUE	
Interactions significant at the 10 per cent level	2	2	1	
Interactions significant at the 5 per cent level	0	0	1	
Item non-response moves the analysed sample only slightly			TRUE	
Liking is measured on a seven-point scale	7	7	1	
Liking runs from 1	1	1	1	
Liking runs to 7	7	7	1	
Linked fate is said to run from 1	1	0	0	paper_internal
Linked fate is said to run to 4	4	3	0	paper_internal
Preview: Spanish ad raises Bush support by about five points	5	4.897	1	
Sampling: Bush respondents who passed the language quiz	1862	1862	1	
Sampling: Latinos who responded to the Bush survey	2866	2866	1	
Sampling: bilinguals who responded to the Vela and Coffman survey	2233	2233	1	
Sampling: nationally representative subjects supplied	2230	2230	1	
Sampling: the same count, restated	1862	1862	1	
Sampling: those who did not pass the language quiz	1344	1344	1	
Sampling: those who passed the language quiz	1681	1681	1	
Six advertisements in all	6	6	1	
Spanish ad raises bilingual support by five percentage points	5	4.876	1	
Table 2 row label: Education(5 levels)	5	5	1	
Table 2 row label: Income(7 levels)	7	7	1	
Table 2 row label: Income(9 levels)	9	9	1	
Table 5 covers all three experiments	3	3	1	
The Republican and Democrat effects do not differ significantly			TRUE	
The Spanish-language survey is said to be in supplemental Appendix C			FALSE	paper_internal
The Vela monolingual effect cannot be distinguished from zero			TRUE	
The Vela point estimate is identical to the Bush one			TRUE	
The calibration supposes a five-point bilingual effect	5	5	1	
The calibration supposes a negative fifteen-point monolingual effect	-15	-15	1	
The interaction coefficients are small and insignificant			TRUE	
The language quiz is said to be in supplemental Appendix D			FALSE	paper_internal
The logit tables are said to be in supplemental Appendix A			FALSE	paper_internal
The simulated electorate runs from 0 per cent bilingual	0	0	1	
The simulated electorate runs to 100 per cent bilingual	100	100	1	
Three candidates produced matched advertisements	3	3	1	
Three randomized survey experiments	3	3	1	
Three separate experiments were conducted	3	3	1	

Table 9: Quantities the article states in a sentence, against the maintained rewrite. *(continued)*

Claim	Paper	Rewrite	Holds	Locus
Twelve interaction terms	12	12	1	
Twelve interaction terms, restated	12	12	1	
Two dependent variables are binary	2	2	1	
Two of three experiments raise bilingual support by about five points			TRUE	
Vela and Coffman experiments: bilinguals recruited	2233	2233	1	
Vela and Coffman experiments: bilinguals who passed the quiz	1681	1681	1	
Vela bilingual advertisement effect	4.9	4.855	1	
Vela bilingual advertisement effect, standard error	2.4	2.39	1	
Vela monolingual advertisement effect	-2	-2.012	1	
Vela monolingual effect on candidate caring	-15.2	-15.2	1	
Vela monolingual effect on candidate caring, standard error	2.5	2.466	1	

Two of these deserve their own note. The appendix claims that the average marginal effects from the logit models "match the OLS models to the second decimal place or better"; across the ten models the largest gap is 1.6e-06, three orders of magnitude tighter than the claim. And the Results section says that "formal tests indicate that item non-response is unlikely to be related to treatment assignment"; none of the deposit's eight scripts runs such a test, and adding one would estimate something the deposit never did, so the claim has no counterpart and its row carries a locus of `archive`. What is checkable is the size of the non-response, which moves the analysed sample by less than one per cent within any candidate-by-sample cell.

11 Maintained rewrite verification

Two full runs of `run_all.R` from clean sessions both exit 0 and produce a byte-identical `maintained/output/`, the three figure PDFs included: `blank_pdf_timestamps()` in `helpers.R` overwrites the `/CreationDate` and `/ModDate` a PDF records when it is written, so the reproduction diff covers every file the pipeline writes rather than all but the figures. Nothing in the pipeline draws a random number, so there is no seed to pin and no sampler change to inherit.

12 R environment

Item	Value
R version	4.6.0
Platform	aarch64-apple-darwin23

Date run	2026-08-10

Table 11: Package versions used for the run behind this report.

Package	Version
estimatr	1.0.6
marginaleffects	0.32.0
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
here	1.0.2